

Nuclear Fuel Performance

NE-533
Spring 2026

MOOSE Notes

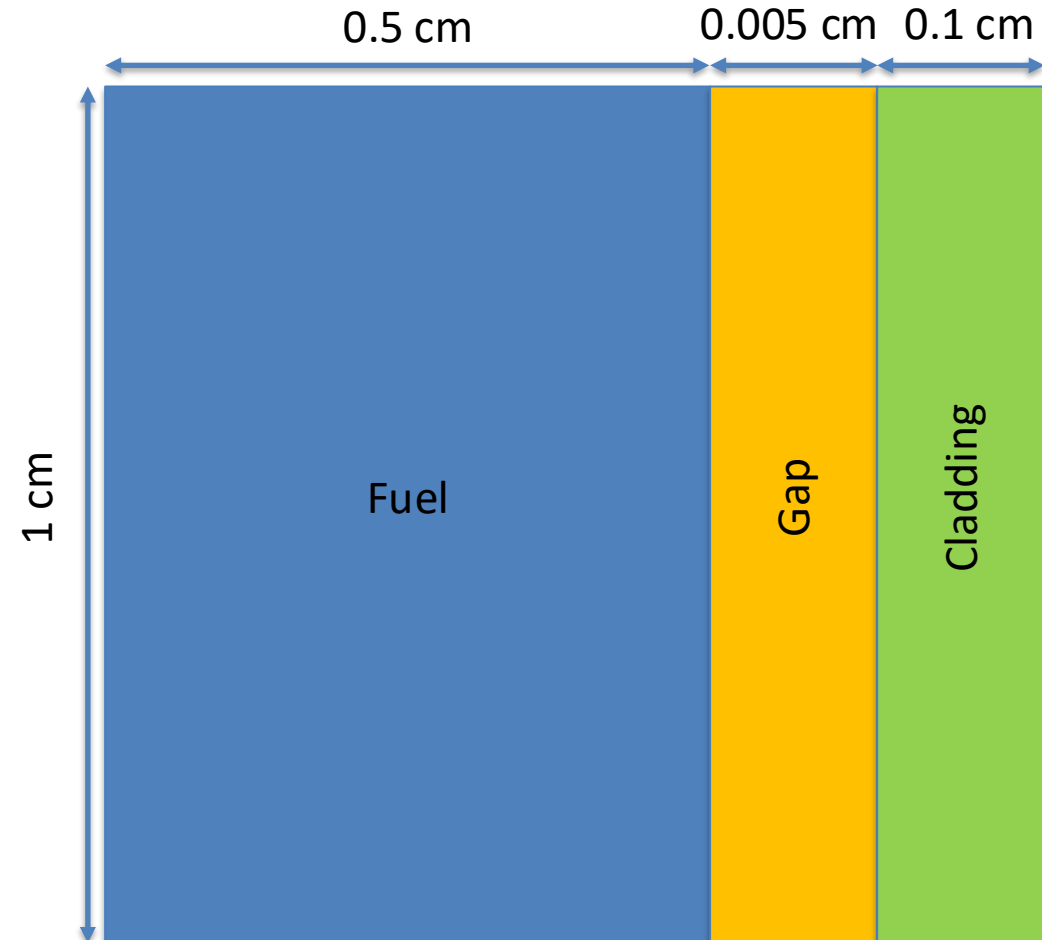
- When you login: enter “groups” in the command line
- Verify that ne533s26 is listed – this should be fixed... but I want you to verify
- Ensure that you have set conda up!
- (base) [bwbeeler@login01 ~]\$
- conda activate /usr/local/usrapps/ne533s26/bwbeeler/moose
- cd /share/ne533s26/\$USER
- #Make your 'input.file'
- mpirun -np 2 /usr/local/usrapps/ne533s26/beeler_moose/beeler_moose-opt -i input.file

MOOSE Project

- Three-part project
- Will upload input and output files to Moodle
- Will upload a written Part 1 report, max of 10 pages (including figures)
- This is an individual project, but some collaboration is encouraged
- Write up with deliverables from Part 1, choice of materials, mesh, details therein, etc.
- Part 1 is due March 3

MOOSE Project Part 1

- Fuel pellet dimensions listed
- This is a 1-D problem, but I want your geometry to be set up in 2-D RZ
- Assume reasonable values for material properties
- Outer cladding temperature is constant: 600 K
- Solve temperature profile for:
 - Steady-state: $Q = 350 \text{ W/cm}^2$
 - Compare against analytical solution
- Use both a constant k and a temperature-dependent k
- Perform a mesh convergence study
- Use transient solver
- Explore two mesh options:
 - explicitly meshing the gap
 - without meshing the gap



Last Time

- Defined strain as a matrix of deformations
- Tensile test stress/strain curves
 - Youngs modulus, yield strength, ultimate tensile strength, fracture strength
- Matrix of elastic constants relating strains to stresses
 - only two unique constants for isotropic materials
- Strain hardening
 - permanent deformation, leading to increased yield strength
 - dislocation pileup, increase in number of barriers
- We developed analytical solutions for the stress in a pressurized cylinder with a thin wall

Develop constitutive relations for thick-walled

- We assume small strains, so the strain is defined with respect to displacement: u

$$\epsilon_{rr} = \frac{\partial u_r}{\partial r}, \quad \epsilon_{\theta\theta} = \frac{u_r}{r}, \quad \epsilon_{zz} = \frac{\partial u_z}{\partial z}, \quad \epsilon_{rz} = \frac{1}{2} \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right)$$

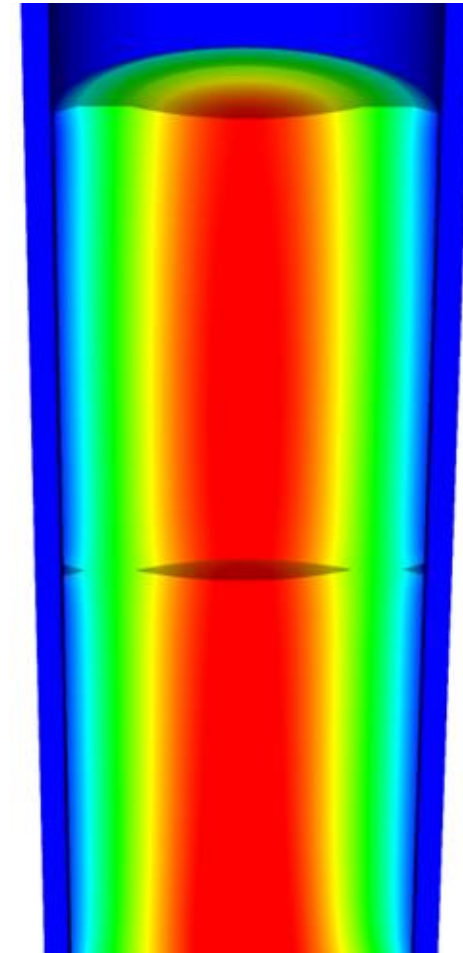
- We assume isotropic material response:

$$\epsilon_{rr} = \frac{1}{E} (\sigma_{rr} - \nu(\sigma_{\theta\theta} + \sigma_{zz}))$$

$$\epsilon_{\theta\theta} = \frac{1}{E} (\sigma_{\theta\theta} - \nu(\sigma_{rr} + \sigma_{zz}))$$

$$\epsilon_{zz} = \frac{1}{E} (\sigma_{zz} - \nu(\sigma_{\theta\theta} + \sigma_{rr}))$$

$$\epsilon_{rz} = \frac{1}{2G} \sigma_{rz}$$



Stress within a pressurized cylinder that has thick walls (radius/thickness < 20)

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} + \frac{\partial \sigma_{rz}}{\partial z} = 0 \quad \frac{1}{r} \frac{\partial (r \sigma_{rz})}{\partial r} + \frac{\partial \sigma_{zz}}{\partial z} = 0$$

- The cylinder as an inner radius R_i and an outer Radius R_o
- We assume there is no shear stress, so $\sigma_{rz} = 0$

- We will begin with the r equation $\frac{\partial \sigma_{rr}}{\partial r} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = 0 \quad \frac{\partial \sigma_{zz}}{\partial z} = 0$

$$\sigma_{\theta\theta} = \sigma_{rr} + r \sigma_{rr,r} \quad \sigma_{\theta\theta,r} = 2 \sigma_{rr,r} + r \sigma_{rr,rr}$$

- Now we need our constitutive law

$$E \epsilon_{\theta\theta,r} = (\sigma_{\theta\theta,r} - \nu(\sigma_{rr,r} + \sigma_{zz,r}))$$

$$E \epsilon_{zz,r} = (\sigma_{zz,r} - \nu(\sigma_{\theta\theta,r} + \sigma_{rr,r})) = 0$$

- We combine these equations to get

$$\begin{aligned} \epsilon_{rr} &= \frac{1}{E} (\sigma_{rr} - \nu(\sigma_{\theta\theta} + \sigma_{zz})) \\ \epsilon_{\theta\theta} &= \frac{1}{E} (\sigma_{\theta\theta} - \nu(\sigma_{rr} + \sigma_{zz})) \\ \epsilon_{zz} &= \frac{1}{E} (\sigma_{zz} - \nu(\sigma_{\theta\theta} + \sigma_{rr})) \end{aligned}$$

$$E \epsilon_{\theta\theta,r} = \sigma_{\theta\theta,r} - \nu(\sigma_{rr,r} + \nu(\sigma_{\theta\theta,r} + \sigma_{rr,r})) \quad E \epsilon_{\theta\theta,r} = (1 - \nu^2) \sigma_{\theta\theta,r} + \nu(1 + \nu) \sigma_{rr,r}$$

$$E(\epsilon_{rr} - \epsilon_{\theta\theta}) = (1 + \nu)(\sigma_{rr} - \sigma_{\theta\theta})$$

Develop equations for the stress within a pressurized cylinder with thick walls

- We need one more relationship from the definition of the strain

- From the previous slide, we have

$$\epsilon_{rr} = \frac{\partial u_r}{\partial r}, \quad \epsilon_{\theta\theta} = \frac{u_r}{r}; \quad \epsilon_{\theta\theta,r} = \frac{1}{r} u_{r,r} - \frac{1}{r^2} u_r = \frac{1}{r} (\epsilon_{rr} - \epsilon_{\theta\theta})$$

- If we combine these, we get

$$E \epsilon_{\theta\theta,r} = (1 - \nu^2) \sigma_{\theta\theta,r} + \nu(1 + \nu) \sigma_{rr,r} \qquad E(\epsilon_{rr} - \epsilon_{\theta\theta}) = (1 + \nu)(\sigma_{rr} - \sigma_{\theta\theta})$$

- Again, from the previous slide

$$(1 - \nu^2) \sigma_{\theta\theta,r} + \nu(1 + \nu) \sigma_{rr,r} = \frac{1}{r} (1 + \nu) (\sigma_{rr} - \sigma_{\theta\theta})$$

- We end up with

$$\sigma_{\theta\theta,r} = 2 \sigma_{rr,r} + r \sigma_{rr,rr} \qquad \sigma_{\theta\theta} = \sigma_{rr} + r \sigma_{rr,r}$$

$$r \sigma_{rr,rr} + 3 \sigma_{rr,r} = 0$$

$$\frac{\partial}{\partial r} \left(r^3 \frac{\partial \sigma_{rr}}{\partial r} \right) = 0$$

We now have an ODE so we can solve to get the stress

- Our boundary conditions are $\sigma_{rr}(R_i) = -p$ and $\sigma_{rr}(R_o) = 0$
- After integrating twice, we get $\frac{\partial}{\partial r} \left(r^3 \frac{\partial \sigma_{rr}}{\partial r} \right) = 0$

- From σ_{rr} , we can get $\sigma_{\theta\theta}$ from
- From the end condition, we determine σ_{zz}

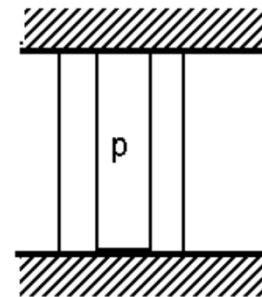
$$\sigma_{rr}(r) = -p \frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1}$$

$$\sigma_{\theta\theta} = \sigma_{rr} + r\sigma_{rr,r} \quad \sigma_{\theta\theta}(r) = p \frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1}$$

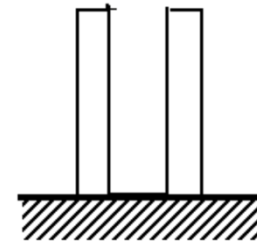
$$(a) \quad \sigma_{zz} = p \frac{2\nu}{(R_o/R)^2 - 1}$$

$$(b) \quad \sigma_{zz} = 0$$

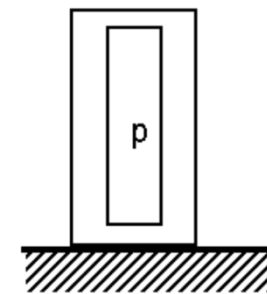
$$(c) \quad \sigma_{zz} = p \frac{1}{(R_o/R)^2 - 1}$$



(a) complete axial restraint



(b) no axial restraint



(c) closed-end tube

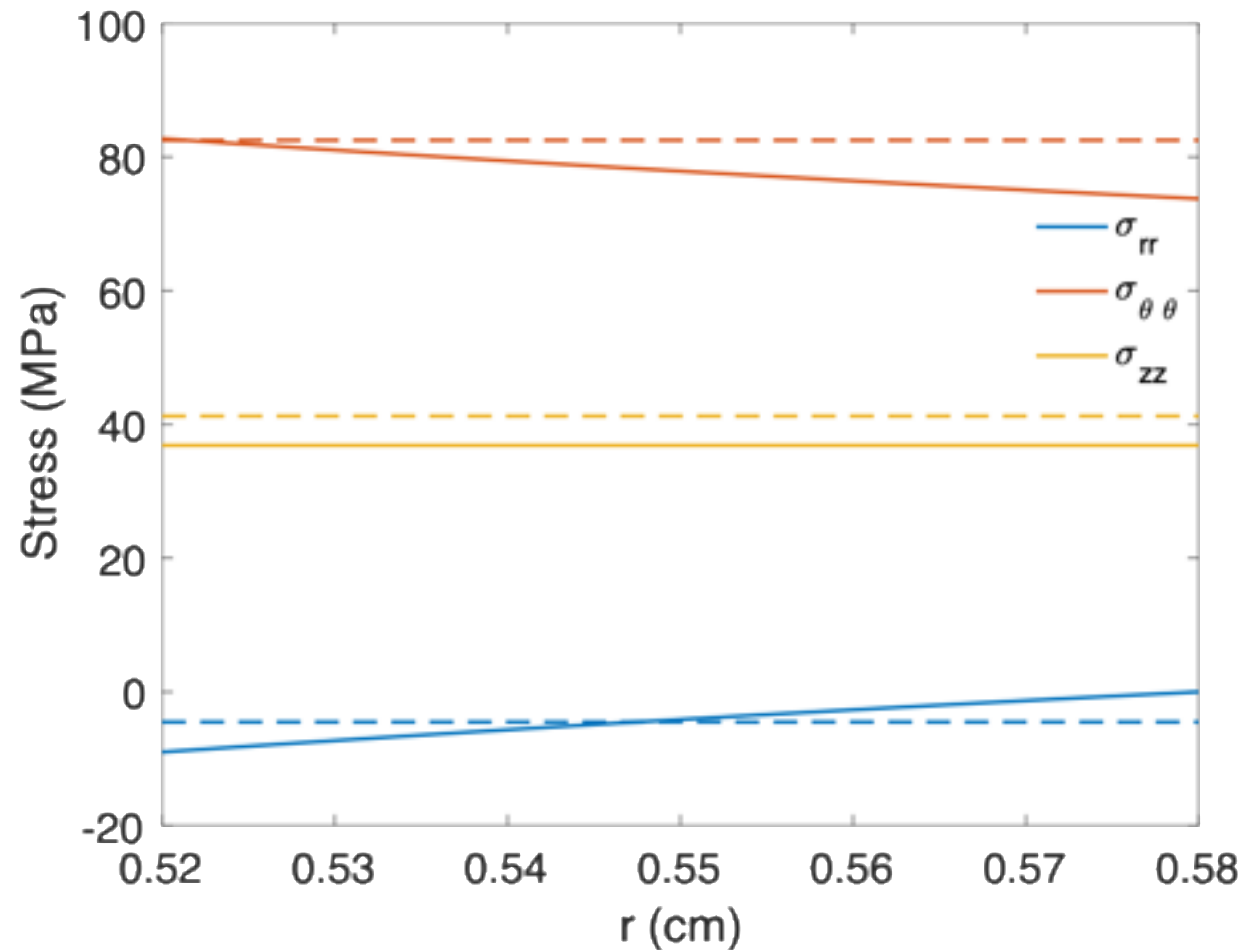
Determine if a fuel rod will exceed its yield stress

- A fuel rod has an initial pressure of 5 MPa that increases to 9 MPa during operation. Will it exceed its yield stress?
 - $R_i = 0.52$, $R_o = 0.58$ cm, $\sigma_y = 381$ MPa

$$\sigma_{rr}(r) = -p \frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1} \quad \sigma_{\theta\theta}(r) = p \frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1} \quad \sigma_{zz} = p \frac{1}{(R_o/R_i)^2 - 1}$$

- Which stress is the largest?
 - $\sigma_{\theta\theta}$
- At what position r will the hoop stress be the largest?
 - $r = R_i = 0.52$ cm
- What is the stress at $p = 5$ MPa?
 - $\sigma_{\theta\theta} = 5 * ((0.58/0.52)^2 + 1) / ((0.58/0.52)^2 - 1) = 46.0$ MPa
- What is the stress at $p = 9$ MPa?
 - $\sigma_{\theta\theta} = 9 * ((0.58/0.52)^2 + 1) / ((0.58/0.52)^2 - 1) = 82.7$ MPa
- Again, we don't even get close to yielding

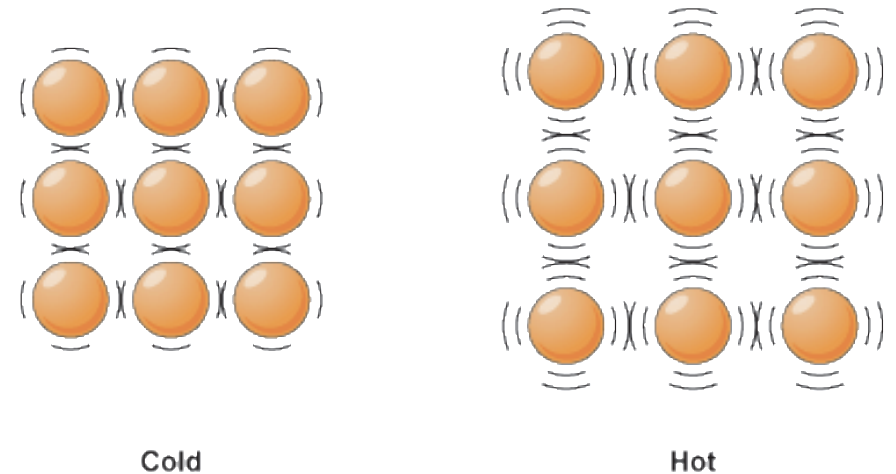
We've now solved this same problem assuming thin and thick-walled cylinders



THERMO-MECHANICS

Thermal Expansion

- As the temperature increases, atoms have larger vibrations, causing the material to expand
- This expansion doesn't raise the energy of the material when unrestrained
- This expansion can be treated as a strain, but as one that doesn't cause stress



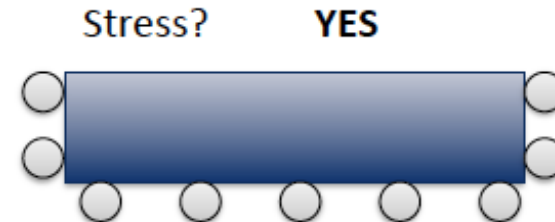
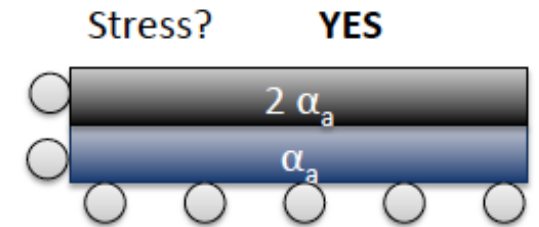
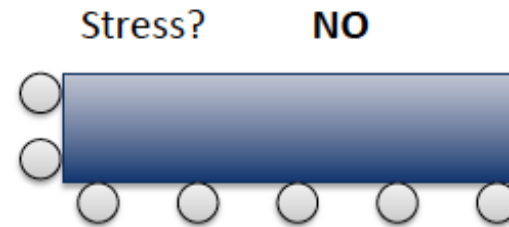
Thermal Expansion

- In isotropic materials, thermal expansion happens equally in all directions and is linear with temperature
- $\epsilon_0 = (T - T_0)\alpha I$
- In this equation
 - T is the current temperature
 - T_0 is the temperature the original size was measured
 - α is the linear thermal expansion coefficient
 - I is the identity tensor

Material	$\alpha (\times 10^{-6} \text{ 1/K})$
Aluminum	24
Copper	17
Steel	13
UO ₂	11
Zircaloy (Axial)	5.5
Zircaloy (radial)	7.1

Thermal Expansion

- Though thermal expansion doesn't directly cause stress, it can still lead to thermal stress



What is the stress in a thin constrained rod of length L when it is heated to ΔT ?

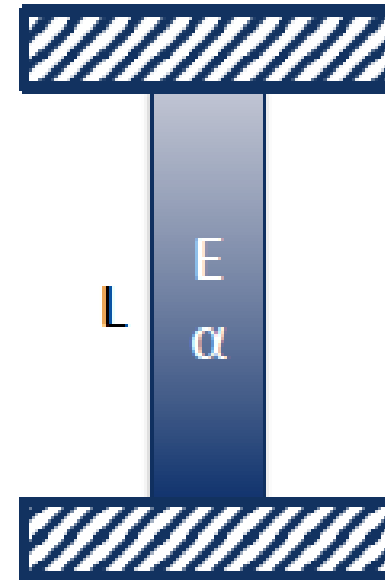
- The rod has a Young's modulus of E and an expansion coefficient of α

$$\epsilon_T = (T - T_0)\alpha \quad \sigma = E (\epsilon - \epsilon_T)$$

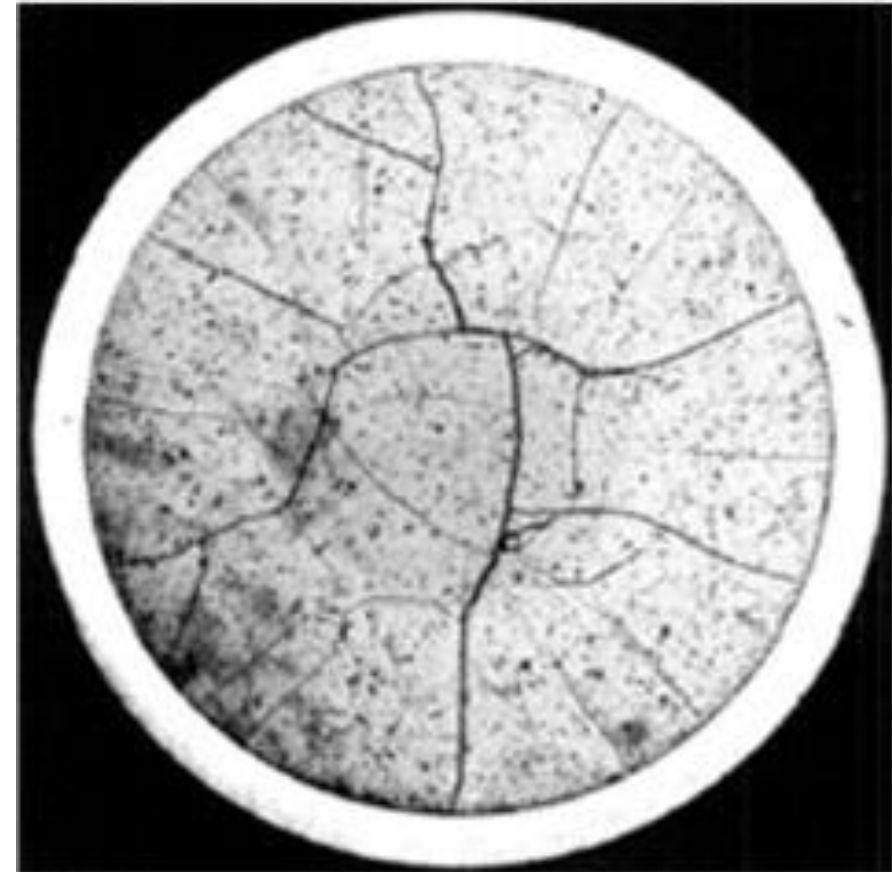
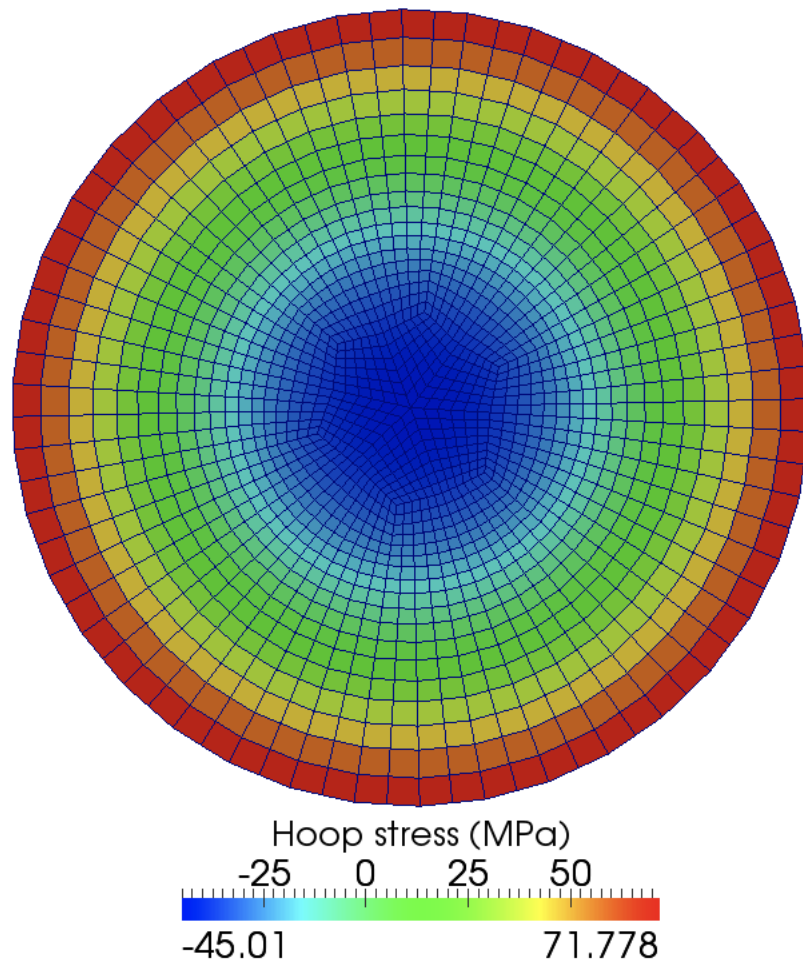
$$\epsilon_0 = (T - T_0)\alpha$$

$$\sigma = E (0 - \Delta T \alpha)$$

$$\sigma = -E\Delta T \alpha$$



The large temperature gradient within a fuel pellet results in large thermal stresses



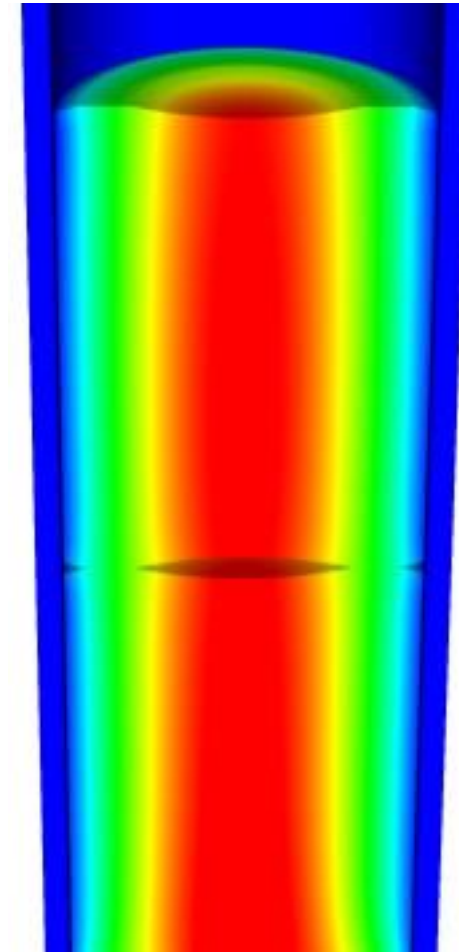
Consider the constitutive relations plus T

- We assume small strains, so the strain is defined as

$$\epsilon_{rr} = \frac{\partial u_r}{\partial r}, \quad \epsilon_{\theta\theta} = \frac{u_r}{r}, \quad \epsilon_{zz} = \frac{\partial u_z}{\partial z}, \quad \epsilon_{rz} = \frac{1}{2} \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right)$$

- We assume isotropic material response, so

$$\begin{aligned} \epsilon_{rr} &= \frac{1}{E} (\sigma_{rr} - \nu(\sigma_{\theta\theta} + \sigma_{zz})) + \alpha\Delta T \\ \epsilon_{\theta\theta} &= \frac{1}{E} (\sigma_{\theta\theta} - \nu(\sigma_{rr} + \sigma_{zz})) + \alpha\Delta T \\ \epsilon_{zz} &= \frac{1}{E} (\sigma_{zz} - \nu(\sigma_{\theta\theta} + \sigma_{rr})) + \alpha\Delta T \\ \epsilon_{rz} &= \frac{1}{2G} \sigma_{rz} \end{aligned}$$



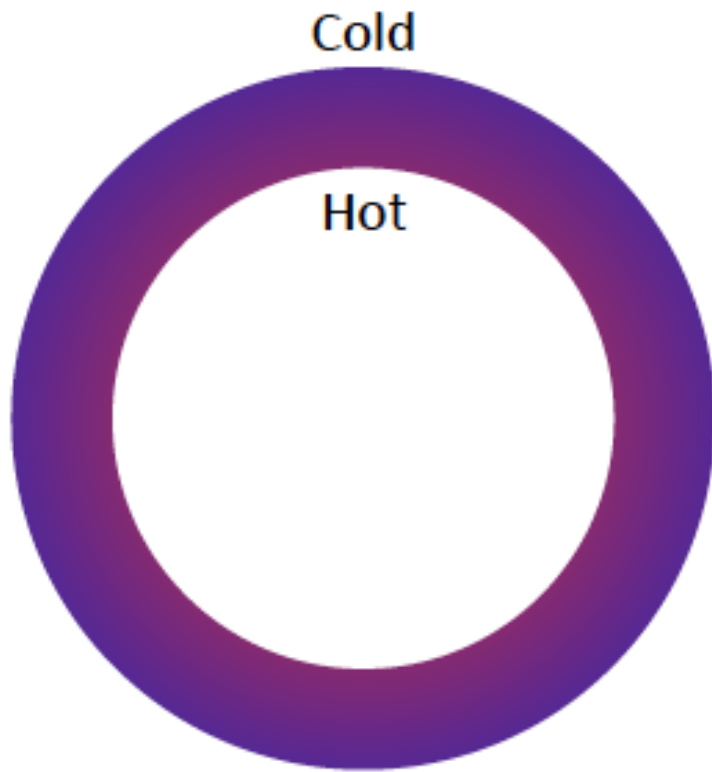
Consider a cylinder with thermal expansion but not pressure

- $\sigma_{rr}(R_i) = \sigma_{rr}(R_o) = 0$
- Similar to the equations we worked through before
- $\frac{1}{r^3} \frac{d}{dr} \left(r^3 \frac{d\sigma_{rr}}{dr} \right) = - \left(\frac{\alpha E}{1-\nu} \right) \frac{1}{r} \frac{dT}{dr}$
- Solving this ODE:

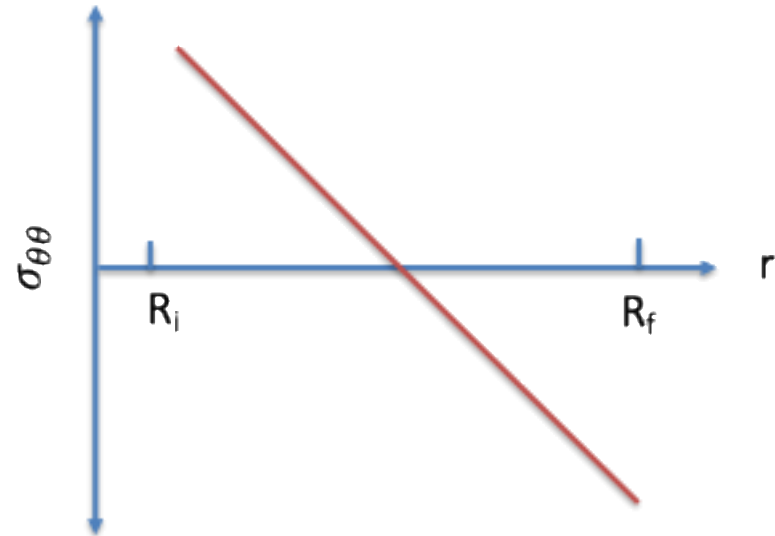
$$\begin{aligned}\sigma_{rr}(r) &= \frac{1}{2} \Delta T \frac{\alpha E}{1-\nu} \left(\frac{r}{R_i} - 1 \right) \left(1 - \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right) \\ \sigma_{\theta\theta}(r) &= \frac{1}{2} \Delta T \frac{\alpha E}{1-\nu} \left(1 - 2 \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right) \\ \sigma_{zz}(r) &= \frac{1}{2} \Delta T \frac{\alpha E}{1-\nu} \left(1 - 2 \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right)\end{aligned}$$



What is the hoop stress in the cladding?



$$\sigma_{\theta\theta}(r) = \frac{1}{2} \Delta T \frac{\alpha E}{1 - \nu} \left(1 - 2 \frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right)$$



Where is hoop stress equal to zero?

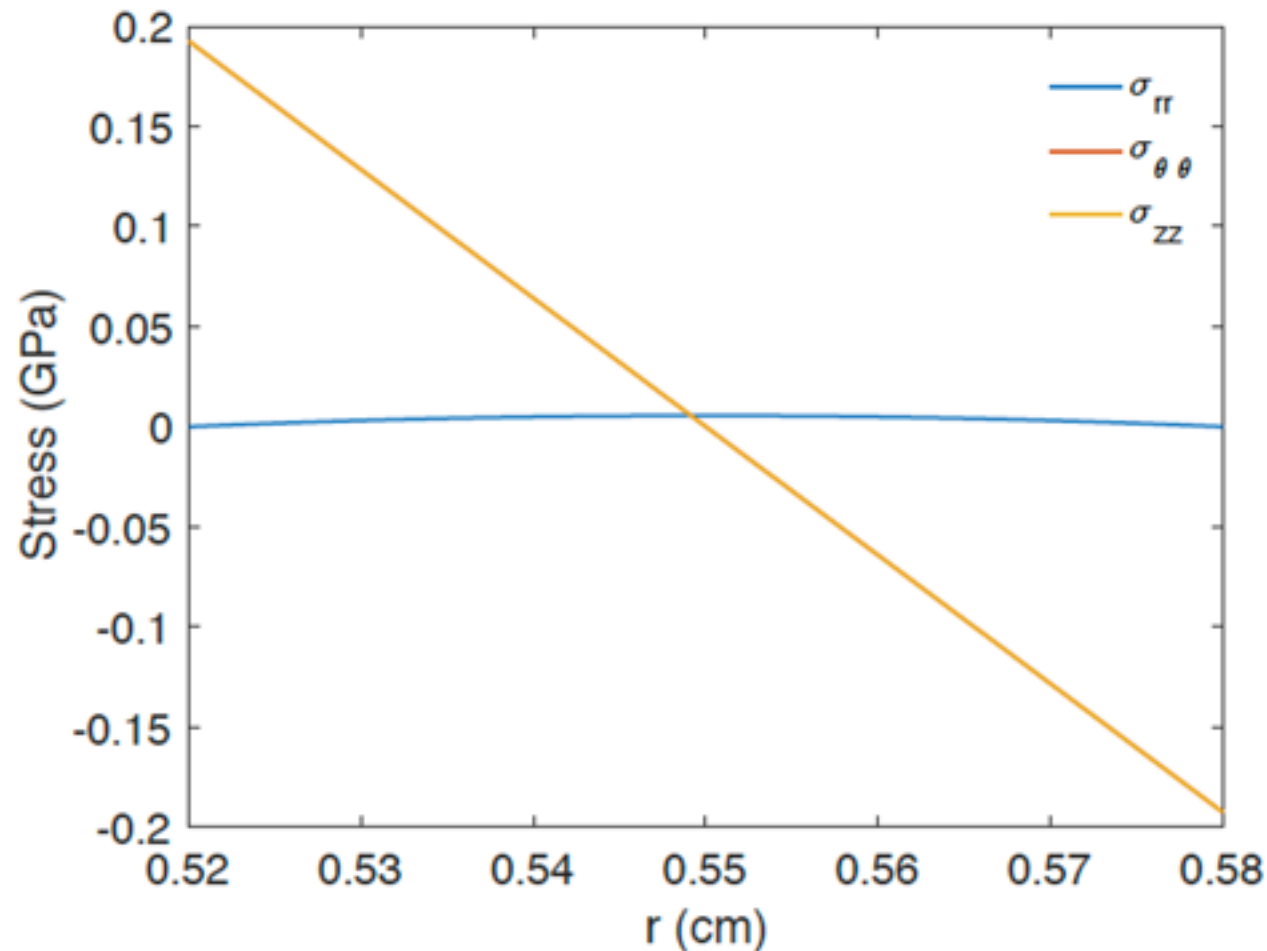
$$\sigma_{\theta\theta}(r) = \frac{1}{2}\Delta T \frac{\alpha E}{1-\nu} \left(1 - 2\frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_{\theta\theta}(r) = \frac{1}{2}\Delta T \frac{\alpha E}{1-\nu} \left(1 - 2\frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right) = 0$$

$$\left(1 - 2\frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) \right) = 0 \quad \longrightarrow \quad 2\frac{R_i}{\delta} \left(\frac{r}{R_i} - 1 \right) = 1 \quad \longrightarrow \quad \left(\frac{r}{R_i} - 1 \right) = \frac{\delta}{2R_i}$$

$$r = \frac{\delta}{2} + R_i$$

The linear temperature gradient across the cladding causes axial thermal stresses



Same approach to the thermal stress in a fuel pellet

- The thermal stress is due to the temperature gradient

$$T(r) = \frac{LHR}{4\pi k} \left(1 - \frac{r^2}{R_f^2} \right) + T_s$$

$$T_0 - T_s = \frac{LHR}{4\pi k}$$

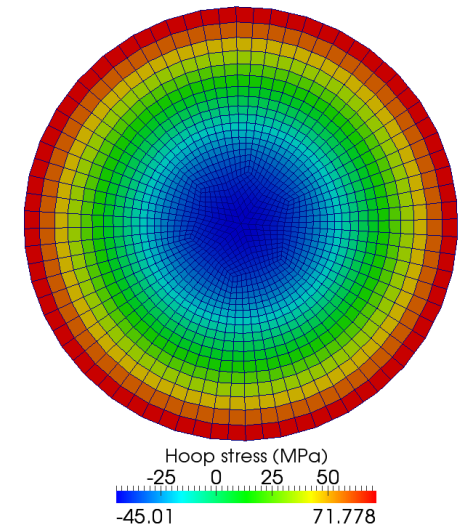
$$T - T_s = (T_0 - T_s) \left(1 - \frac{r^2}{R_f^2} \right)$$

$$\frac{\partial T}{\partial r} = 2(T_0 - T_s) \left(\frac{r}{R_f^2} \right) \quad \frac{1}{r^3} \frac{d}{dr} \left(r^3 \frac{d\sigma_{rr}}{dr} \right) = - \left(\frac{\alpha E}{1 - \nu} \right) \frac{1}{r} \frac{dT}{dr}$$

$$\eta = \frac{r}{R_f}$$

$$\sigma^* = \frac{\alpha E (T_0 - T_s)}{4(1 - \nu)}$$

$$\frac{d}{d\eta} \left(\eta^3 \frac{d\sigma_{rr}}{d\eta} \right) = 8\sigma^* \eta^3$$



Solve this stress ODE

- The boundary conditions are:

$$\frac{d\sigma_{rr}}{d\eta} = 0 \text{ at } \eta = 0$$

$$\sigma_{rr} = 0 \text{ at } \eta = 1$$

- Once we solve it, we obtain
- Then we can solve the hoop stress
- The axial stress is more complicated to obtain, but you end up with

$$\frac{d}{d\eta} \left(\eta^3 \frac{d\sigma_{rr}}{d\eta} \right) = 8\sigma^* \eta^3 \quad \sigma^* = \frac{\alpha E (T_0 - T_s)}{4(1 - \nu)}$$

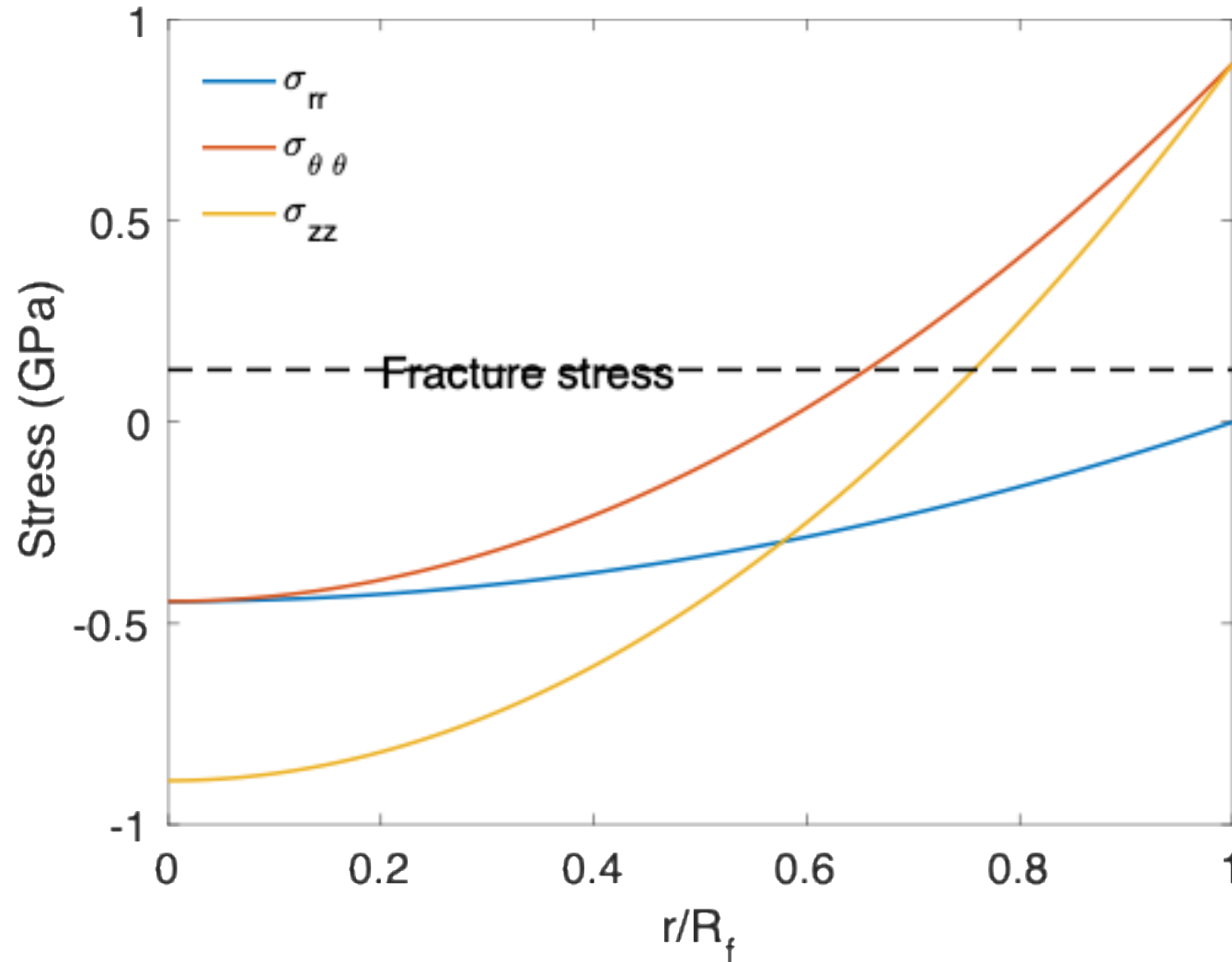
$$\sigma_{rr}(\eta) = -\sigma^*(1 - \eta^2)$$

$$\sigma_{\theta\theta} = \sigma_{rr} + r\sigma_{rr,r}$$

$$\sigma_{\theta\theta}(\eta) = -\sigma^*(1 - 3\eta^2)$$

$$\sigma_{zz}(\eta) = -2\sigma^*(1 - 2\eta^2)$$

The fuel temperature gradient causes large thermal stresses



Example

How far do fuel cracks extend?

$$\sigma_{\theta\theta}(\eta) = -\sigma^*(1 - 3\eta^2) \quad \sigma^* = \frac{\alpha E(T_0 - T_s)}{4(1 - \nu)} \quad \eta = \frac{r}{R_f}$$

- $E = 200 \text{ GPa}$, $\nu = 0.345$, $\alpha = 11.0\text{e-}6 \text{ 1/K}$, $\sigma_{\text{fr}} = 130 \text{ MPa}$, $\Delta T = 550 \text{ K}$
- Solve for η
 - $-\sigma_{\text{fr}} / \sigma^* = 1 - 3 \eta^2$
 - $3 \eta^2 = 1 + \sigma_{\text{fr}} / \sigma^*$
 - $\eta = ((1 + \sigma_{\text{fr}} / \sigma^*) / 3)^{1/2}$
- $\sigma^* = 11.0\text{e-}6 * 200 * 550 / (4 * (1 - 0.345)) = 461.8 \text{ MPa}$
- $\eta = \text{sqrt}((1 + 130/461.8) / 3) = 0.65$

THERMAL EXPANSION

The gap changes as a function of time

- Both the pellet and the cladding swell

$$\Delta\delta_{gap} = \delta_{gap} - \delta_{gap}^0 \quad \Delta\delta_{gap} = \Delta\bar{R}_C - \Delta R_f$$

$$\frac{\Delta R_C}{\bar{R}_C} = \alpha_C (\bar{T}_C - T_{fab}) \quad \frac{\Delta R_f}{R_f} = \alpha_F (\bar{T}_f - T_{fab})$$

$$\Delta\delta_{gap} = \bar{R}_c \alpha_C (\bar{T}_C - T_{fab}) - \bar{R}_f \alpha_f (\bar{T}_f - T_{fab})$$

- But, as the gap decreases, the temperature changes, which again makes the gap change
- The solution using the analytical equations is iterative, due to the dependence of the gap size and temperature

Example

This calculation is repeated until the gap width stops changing significantly

- The change in the gap does NOT affect the coolant or outer cladding temperatures, just the gap and fuel temperatures
- Does fuel centerline temp go up or down with correction?

Iteration	δ_{gap} (cm)	T_s (K)	T_o (K)
0	0.003	701	1232
1	0.00066	644	1174
2	0.00097	652	1182
3	0.00094	651	1181
4	0.00094	651	1181

